

Approximating spatial processes with too many knots degrades predictive accuracy of SPDE models

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Abstract

1. Spatial and spatiotemporal models are seeing increasingly used in ecology and related fields to model latent distributions. Applications relying on these methods have used them to track population change over time, assess changing species distributions, and model spatial population processes (predation, habitat use, fishing impacts, etc).
2. Computational advances over the last decade have allowed for the rapid estimation of spatial approximations to high dimensional spatial surfaces (100s or more georeferenced points), and these methods have been accessible in software implementing the Integrated Nested Laplace Approximation (INLA). One of the most critical decisions for building INLA-based models is how to construct the spatial approximation (“mesh”). Little guidance has been established in constructing these approximations, with many previous applications suggesting that higher resolution meshes result in better performance. Using a publicly available dataset on commercially fished species on the west coast of the USA, we develop spatial models of species occurrence and catch rates for 4 species. We conduct a series of sensitivity analyses to understand how mesh resolution affects the predictive accuracy of models predicting species occurrences and catch rates.
3. Our results highlight that for both presence-absence (binomial) and positive (Gamma) models, the

finest scale spatial approximations considered overfit the data and resulted in decreased out of sample predictive performance. For each of the species and models considered, there appeared to be an optimal range of vertices (knots) used in constructing the spatial approximation. These results held across a range of spatial cross-validation sampling schemes, including folds selected in 1-dimensional latitudinal blocks, and the use of 2-dimensional spatial blocks.

4. Results from our work highlight that practitioners should not assume that the performance of INLA-based models gets better with higher resolution. Instead, each dataset likely has a range of resolutions that is optimal, and these optimal ranges may be identified using spatial cross-validation approaches, such as the sampling procedures used in our analysis. Selecting models with the appropriate spatial complexity is expected to improve parameter estimates and derived quantities (estimates of population abundance, distribution, etc.).

Key words

Spatial, spatiotemporal, Integrated Nested Laplace Approximation (INLA), Gaussian random field, mesh, predictive accuracy

Introduction

Recent computational advances have allowed for the rapid estimation of large spatial and spatiotemporal models in ecology and related fields (Banerjee & Fuentes 2012). Motivating questions for these applications to large georeferenced datasets including tracking abundance in time and space, identifying environmental predictors of occurrence, modeling disease spread, or quantifying range shifts in distribution (Ross *et al.* 2012; Miller *et al.* 2013; Thorson & Barnett 2017).

There are a number of statistical approaches for developing models with spatial components; examples include generalized additive models (GAMs) (Murase *et al.* 2009; Forney *et al.* 2012), non-parametric approaches (Mutanga *et al.* 2012), and applications of Gaussian random fields (Latimer *et al.* 2009; Datta *et al.* 2016). Estimating these models can be computationally challenging for large spatial datasets (Banerjee *et al.* 2004), necessitating the use of approximations or smoothing. One solution to this problem is the Stochastic Partial Differentiation Equation (SPDE) approach (Lindgren *et al.* 2011). The SPDE method uses Gaussian Markov Random Fields (GMRF) to model the conditional distributions of values as only dependent on neighboring points; thus, rather than estimating a full rank covariance method to describe the spatial field, a sparse precision matrix can be estimated (Lindgren *et al.* 2011). Like other approximations, the SPDE approach can be seen as a form of smoothing (Yue *et al.* 2014). (Miller *et al.* 2020) provide

an excellent summary of the recent SPDE literature, and also show the link between these approaches and smoothing splines.

Advances by (Rue *et al.* 2009) have enabled the SPDE approach to be implemented with the Integrated Nested Laplace Approximation (INLA) as a fast and efficient approximation to estimating high-dimensional Gaussian random fields. Rather than using MCMC to estimate the posterior distribution of parameters, INLA approximates the posterior with the Laplace approximation. These models have been extended to include the ability to fit geographic barriers, such as islands or lakes, anisotropic covariance functions, and non-Gaussian data (Bakka *et al.* 2018). In addition to estimating Bayesian posterior distributions (Lindgren & Rue 2015), INLA has been integrated with Template Model Builder (TMB; (Kristensen *et al.* 2016; Osgood-Zimmerman & Wakefield 2021)) to enable auto-differentiation and has been adopted in many spatial fisheries applications around the world (Anderson *et al.* 2019a; Thorson 2019b; e.g., Johnson *et al.* 2019).

Constructing spatiotemporal models with INLA or related software involves many of the same considerations as with simpler non-spatial models. For example, modelers frequently ask: “Which covariates are most important?” and “Does my model do a satisfactory job of making predictions?” Spatial or spatiotemporal models may include additional considerations, such as specifying the form of spatiotemporal variation (Anderson & Ward 2019). One of the more complicated and important decisions—particularly for models implementing the SPDE–INLA approach—lies in specifying the dimensionality of the latent spatial fields (Lindgren & Rue 2015). Unlike Generalized Additive Models (GAMs), which often penalize smooth terms (Wood 2003), there are not explicit penalties in overfitting with INLA.

One of the first steps in fitting a model with INLA involves choosing the appropriate resolution of the spatial field using a “mesh” (a map consisting of small triangles, e.g., Figure 1). By estimating values of the spatial field at the vertices (or “knots”), the spatial field may be triangulated to coordinates where data are observed or predictions are made (Lindgren *et al.* 2011). There are a number of decision points to be made in constructing the mesh, including whether to include a boundary, whether the boundary is regular, the maximum edge of any triangle (in the core or boundary area), and the minimum angles allowed by the triangles (Gómez-Rubio 2020). Each of these decisions can impact the density and number of small triangles that make up the mesh—and in turn, the accuracy of predictions. Though the mesh construction is important, this step has been omitted from best practices guides using an the SPDE–INLA approach, and we are only aware of one sensitivity analysis examining the importance of mesh construction (Righetto *et al.* 2020). While many previous applications and guidance around INLA have suggested that increasing the resolution of the mesh improves predictive accuracy (Thorson 2019a; Godana *et al.* 2019; Williamson *et al.*

2021; Wilson 2021), a much smaller number of studies have noted that finer meshes don't always translate into better performance (Huang *et al.* 2017).

The objective of this paper is to examine whether higher resolution spatial approximations in INLA translate to better predictive accuracy, and provide general recommendations for identifying models with good out of sample predictive performance. Using a series of analyses on a real-world case study of groundfishes in the Northeast Pacific Ocean, we evaluate the out of sample predictive performance as a function of mesh resolution (number of knots). We further evaluate these results across a range of cross-validation sampling schemes (number of spatial blocks and folds), including an extreme case of leave-one-out cross-validation (LOOCV). All data for our analysis are publicly available, and code and data to reproduce our analysis is available on Github (XXXX) and Zenodo (on publication).

Methods

Data

To evaluate the effect of mesh resolution on predictive ability of INLA-based models, we use a publicly available dataset of catch rates from scientific surveys in the Northeast Pacific Ocean as a motivating example (<https://www.nwfsc.noaa.gov/data>). The West Coast Bottom Trawl Survey (WCBTS) data represents a fishery-independent survey collected annually over a 19-year period, 2003-2021, covering the California Current ecosystem on the west coast of the USA (Keller *et al.* 2017). The survey is designed to estimate the abundance, sizes, and age composition of commercially and recreationally targeted groundfish species found in near-bottom habitats. The fundamental methods of the survey, sampling intensity, gear, seasonal and spatial coverage, and random stratified design have remained relatively constant. Because of the large number of model runs needed to evaluate our sensitivity analyses, we restricted our analysis to only use data from 2018 ($n = 702$ observations). Though hundreds of species were encountered in the survey, we further restricted our analysis to four species that are of interest to fisheries, but also represent a range of occurrences: Dover Sole (*Microstomus pacificus*, encountered in 593 tows), Petrale Sole (*Eopsetta jordani*, 291 tows), Lingcod (*Ophiodon elongatus*, 225 tows), and Darkblotched Rockfish (*Sebastes crameri*, 105 tows). The overall mean distance between tow locations in the WCBTS was 676.7km ($\tilde{x} = 603.1$ km).

Modeling

Because of the presence of zeros and positive continuous values in fish survey datasets such as the WCBTS data, conventional approaches to analyzing these data have used a delta or hurdle model to separately model occurrence and positive catches (Thorson 2019a; e.g., Johnson *et al.* 2019). We adopt a similar approach

in our modeling, and apply two sub-models to the data; we use a binomial distribution with a logit link to model species occurrence, and a Gamma distribution with a log link to model positive catch rates (kg per km²). For simplicity and consistency with fisheries assessments using these data (@ Johnson *et al.* 2019), we fit models estimating an intercept and spatial field throughout, e.g.

$$g(u_s) = B_0 + \omega_s$$

where ω_s is the estimated spatial field, B_0 is the intercept, u_s is the expected value for point in space s , and $g()$ is an inverse link function.

All of the meshes in our analysis used a non-convex boundary and offset based on 5% of the data (Figure 1). We fixed the maximum edge of triangles in the mesh at 100km in the core area and 500km closer to the boundary (Figure 1). Throughout our analyses, we adjusted the mesh resolution by varying the cutoff distance (representing the distance where two points are treated as a single vertex) because the cutoff distance has been shown to have the greatest impact on measures of model performance from mesh construction (Righetto *et al.* 2020). We fit all models using R-INLA (Lindgren & Rue 2015; R Core Team 2021) and R 4.1.0 (R Core Team 2021) and specifically the ‘inlabru’ R package, which is a convenient wrapper for the INLA software (Bachl *et al.* 2019). We used penalized complexity priors on the Matérn covariance parameters (Simpson *et al.* 2015; Osgood-Zimmerman & Wakefield 2021). Specifically, priors on the Matérn spatial range (κ) and marginal standard deviation (σ_ω) are constructed as a function of the tail probabilities (Simpson *et al.* 2017; Fuglstad *et al.* 2018). We assigned priors so that $Pr(\kappa < 20) = 0.05$ and $Pr(\sigma_\omega > 5) = 0.05$. The remaining priors were on the global intercept, $B_0 \sim N(0, \infty)$ and for the Gamma sub-model, the default log-Gamma prior was assigned to the precision $\log(\tau)$ *Gamma*(1, 0.01).

Sensitivities

As a preliminary evaluation of whether the spatial resolution of INLA-based models can result in overfitting, we evaluated 20 cutoff values ranging from 3–120km for each species and hurdle sub-model included in our analysis. There is an inverse relationship between cutoff values and mesh resolution or knots; while the meshes varied slightly by species, this range of cutoff produced meshed that ranged in size from 75–1348 knots. High-resolution meshes with more knots than observations were included as an upper extreme, though many applications of INLA to similar fisheries survey data have used 500–600 knots (e.g., Thorson 2019a; Tolimieri *et al.* 2020).

To evaluate model performance, we conducted 10-fold cross-validation. Though not a 1-dimensional dataset, the strong latitudinal gradient has driven spatial stratification in previous analyses (Keller *et al.* 2017) (Figure 1). We a priori divided the survey domain into 10 spatial blocks by latitude quantiles with approximately the same number of points in each box (the latitudinal mean distance of each box was 178.1km, $\sigma=56.2$). Each model was run using `inlabru` (@ Bachl *et al.* 2019), and the fit to each set of training data was used to predict observations in the held out block. The predictive density was calculated using the binomial or Gamma densities, and total predictive densities were calculated by summing across folds. To contrast the out of sample predictive performance with fit to the training data, we also calculated the predictive density of the training data, separately for each fold.

As a second sensitivity, we examined whether the choice of cross-validation sampling may influence inference about predictive ability. Recent work has demonstrated that for spatially structured data, creating random folds that ignore spatial autocorrelation may bias results (Valavi *et al.* 2019). Because of the added dimensions, we restricted this sensitivity to a single sub-model (Dover Sole presence-absence). The performance of cross-validation was compared across a range of mesh resolutions (cutoff values ranging from 3–120, or 75–1348 knots), number of spatial blocks (19–218) and number of folds used for cross-validation (10, 40). Selection of spatial blocks was done using the ‘blockCV’ package in R (Valavi *et al.* 2019) with varying distance bands between test and training sets (25–150km). These bands are used to create buffers around test data, with larger bands creating more separation between test and training data.

As a third sensitivity, we examined the robustness of the cross-validation results to the choice of hyperparameters for the PC priors (Matérn κ and σ_ω). Because of the dimensionality of parameters to explore, we continued to focus on the Dover Sole presence-absence sub-model. The PC prior on σ_ω is to specify the threshold and upper tail probability, $Pr(\sigma_\omega > \sigma_0) = \theta_\sigma$. In addition to the original values ($\sigma_0 = 5$, $\theta_\sigma = 0.05$), we examined alternative values of $\sigma_0 = (1, 10)$ and $\theta_\sigma = c(0.01, 0.1, 0.4)$. In contrast, the PC prior on κ specifies the lower threshold and tail probability, $Pr(\kappa > \kappa_0) = \theta_\kappa$. We varied κ_0 (5, 20, 40) and θ_κ (0.01, 0.1, 0.4). For each combination of PC priors, we calculated the out of sample predictive density, using 10-fold cross-validation and the same latitudinal bands as in our first sensitivity analysis.

To further validate the inference from the spatial cross-validation, we examined the performance of alternative measures of out of sample predictive performance. We focused on the conditional predictive ordinate (CPO, (Pettit 1990)), which is an approximation to the leave-one-out cross-validation score (LOOCV). Ideally, we would use cross-validation to evaluate the out of sample predictive accuracy of a model. The total predictive log density can be represented as the sum of individual log posterior probabilities, $\sum_{i=1}^n \log[P(Y_i|Y_{\neq i})]$, where $P(Y_i|Y_{\neq i})$ is the probability of test data i . Because conducting true leave-one-out cross-validation may

be prohibitive for large datasets, INLA uses CPO to approximate the out of sample posterior probability for each observation. Previous research has shown that the CPO approximation performs similarly to MCMC cross-validation (Held *et al.* 2010), yet it is less clear whether CPO is similar to true cross-validation in comparing mesh resolution. Focusing on the presence-absence model of Dover Sole (Figure 2), we first calculated the total CPO score across a range of cutoff values. For comparison, we then calculated the true predictive density using LOOCV across a similar range of cutoffs (fitting the INLA model 702 times for each value of the range).

As a final sensitivity, we compared parameter estimation using INLA to estimation of the same models using TMB (Kristensen *et al.* 2016). Specifically, we performed estimation using the R package ‘sdmTMB’ (Anderson *et al.* 2019b), which uses the INLA software to create meshes, but takes advantage of autodifferentiation in TMB for faster parameter estimation.

Results

Our 10-fold cross-validation with the binomial and Gamma models across a range of species and mesh cutoff values demonstrated that after some optimal spatial cutoff of triangles in the mesh, predictive accuracy of the spatial models deteriorated as mesh resolution was increased. Binomial models of Darkblotched Rockfish (Fig. 2) and Gamma models for all species also supported a decline in performance with the coarsest mesh resolutions (Fig. S1). The binomial sub-model suggested that the optimal number of knots was 200–250 (Fig. 2), and the positive Gamma sub-model indicated even smaller knot values had nearly equivalent out of sample predictive performance (Fig. S1). In contrast to the decline in out of sample performance with high mesh resolution, we found the opposite to be true for in sample data; for both the binomial and Gamma models, predictive performance generally increased with the number of knots (Fig. 2, Fig. S1).

Focusing on the Dover Sole presence-absence model, we examined sensitivity to spatial sampling and choice of penalized complexity parameters for the Matérn covariance function. To examine effects of spatial sampling, we adjusted the buffer separating test and training dataset (Fig. 3) and adjusted the number of folds used for cross-validation (10, 40, Fig. S3). There is an inverse relationship between the buffer size and number of blocks, with large buffers (125km, 150km) corresponding to small numbers of blocks for this dataset (25, 19). With these largest buffers, there appears to be a decline in predictive performance at small numbers of knots (Fig. 3, Fig. S3), but with large numbers of spatial blocks (> 50), the coarsest meshes have similar performance as models with moderate numbers of knots. Varying the hyperparameters of the PC priors for the Matérn κ and σ_ω similarly demonstrated that the predictive performance of these models declined across all values (Fig. SX).

Because k-fold cross-validation requires significant computation time, approximations like CPO may be more convenient ways to measure predictive accuracy of spatiotemporal models. Though conventionally used to compare models with alternative fixed effects, we compared CPO approximations to true LOOCV, as an extreme case of cross-validation. Both CPO and LOOCV suggested that too many knots can result in diminished predictive accuracy (Figure 4). The biggest difference in these metrics was that the LOOCV suggested that more than 400 knots resulted in decreased performance, while CPO indicated a performance decline after 675 knots (Figure 4).

Finally, we examined the effects of increasing mesh resolution on estimated model parameters. We found little evidence of patterns between mesh resolution, estimated Matérn parameters (κ , σ_ω), and predictive log densities. While the estimated range (κ) generally declined with large numbers of knots (Fig. 5), there are less obvious patterns in the marginal standard deviation σ_ω (Fig. 5). The estimated fixed effect B_0 representing the mean of the spatial field generally declined with increasing mesh resolution (Figure S2), because as the mesh increases in complexity and explains more variation in observed data, the intercept is expected to shrink toward the mean.

TODO (WORK IN SDMTMB COMPARISON, FIGURE FROM SEAN?)

Discussion

Spatial and spatiotemporal models are becoming increasingly used in ecology, fisheries, and related subfields, in particular to address questions related to changing species distributions in time and space. High dimensional datasets with hundreds of observations introduce computational challenges, but complicated spatial structure can be efficiently approximated using the SPDE approach (Miller *et al.* 2020). Combined with the INLA software for estimation (Rue *et al.* 2009), these tools have become widely accessible. Though a number of examples and best practices papers have been written to demonstrate how to construct models or specify hyperparameters (Lindgren & Rue 2015; Gómez-Rubio 2020), advice on specifying the mesh to approximate spatial processes can be confusing. Identifying the spatial complexity introduces new computational challenges for analysts, but is important for improving predictions and derived quantities of interest (indices of population abundance, area occupied, centre of gravity, etc.).

Using a publicly available dataset of groundfishes in the NE Pacific Ocean, we use a number of sensitivity analyses to demonstrate an important tradeoff between mesh resolution and the predictive accuracy of spatial models using the SPDE-INLA approach. As expected, models with higher resolution meshes (more knots or vertices) are more flexible and can better explain variation in data used to fit the model. For applications

interested in identifying models with good predictive ability for future data (either in space or time), we demonstrate that models with the finest spatial approximations have deteriorating predictive performance for out of sample data, when compared to coarser models. Declines in out of sample predictive performance with too many knots appears robust to (1) how data are spatially sampled for cross-validation, (2) the specific penalized complexity priors used for the Matern function, and (3) whether estimation is done using INLA or maximum likelihood with TMB. Datasets like the groundfish case study used here ($n = 702$) support meshes with ~ 300 or fewer knots, though some of the models supported even fewer knots.

While there are a number of alternative ways to measure model performance, we recommend using k-fold cross-validation to explore predictive accuracy across a range of meshes. Alternative measures of model fit exist, including the conditional predictive ordinate (CPO) (Pettit 1990). While the CPO may approximate LOOCV for some datasets, we found a mismatch between CPO and LOOCV, with CPO supporting meshes with very large numbers of knots and true LOOCV supporting fewer. Other tools like AIC have also been used to discriminate among these types of models, however AIC is known to be problematic when used as a metric to compare mixed effects models (Harrison *et al.* 2018) or as a tool to estimate spatial complexity (Velasco & González-Salazar 2019).

Comparing the performance of models with alternative spatial complexity obviously introduces additional computational burdens, both in performing cross-validation, and comparing models with different numbers of knots. This challenge is not unique to models using the INLA–SPDE approach; selecting both the location and number of knots is a general problem for knot-based spatial models. Other examples of spatial statistics models that have encountered this issue include machine learning methods (reviewed in (Banerjee *et al.* 2012)), Gaussian Process models (Garton *et al.* 2020) and P-splines (Ruppert 2002). Identifying a range of knot values that yield good predictive performance will be specific to individual datasets (e.g. magnitude of observation error, spatial autocorrelation, number of observations, type of response). The optimal mesh resolution for one species may be informative for other species from the same dataset, potentially reducing the computational overhead.

3. Rules of thumb? This is tricky because it’s somewhat dataset specific. The blockCV MEE paper suggests defining spatial blocks based on residual spatial autocorrelation – which involves fitting the model, so somewhat circular [and tools like calculating the variogram are made difficult by non-normal response data]. Most advanced GP approaches include knot selection as a parameter, or do an adaptive grid search

The number of knots needed to generate a model with high predictive accuracy depends on the other

features of the model. Some of this is based on data (e.g. homogenous spp like Dover sole vs patchily distributed rockfish), but a lot of this is dependent on how much variation is parsed into observation error. High obs errors will result in flatter spatial fields, which will require fewer knots.

4. Future research:

- is this a feature of triangulation with inla? would we get better estimates from a higher dimensional GP model?

– both bias and precision, fig s2 - don't use AIC - identifying the appropriate spatial complexity is important for

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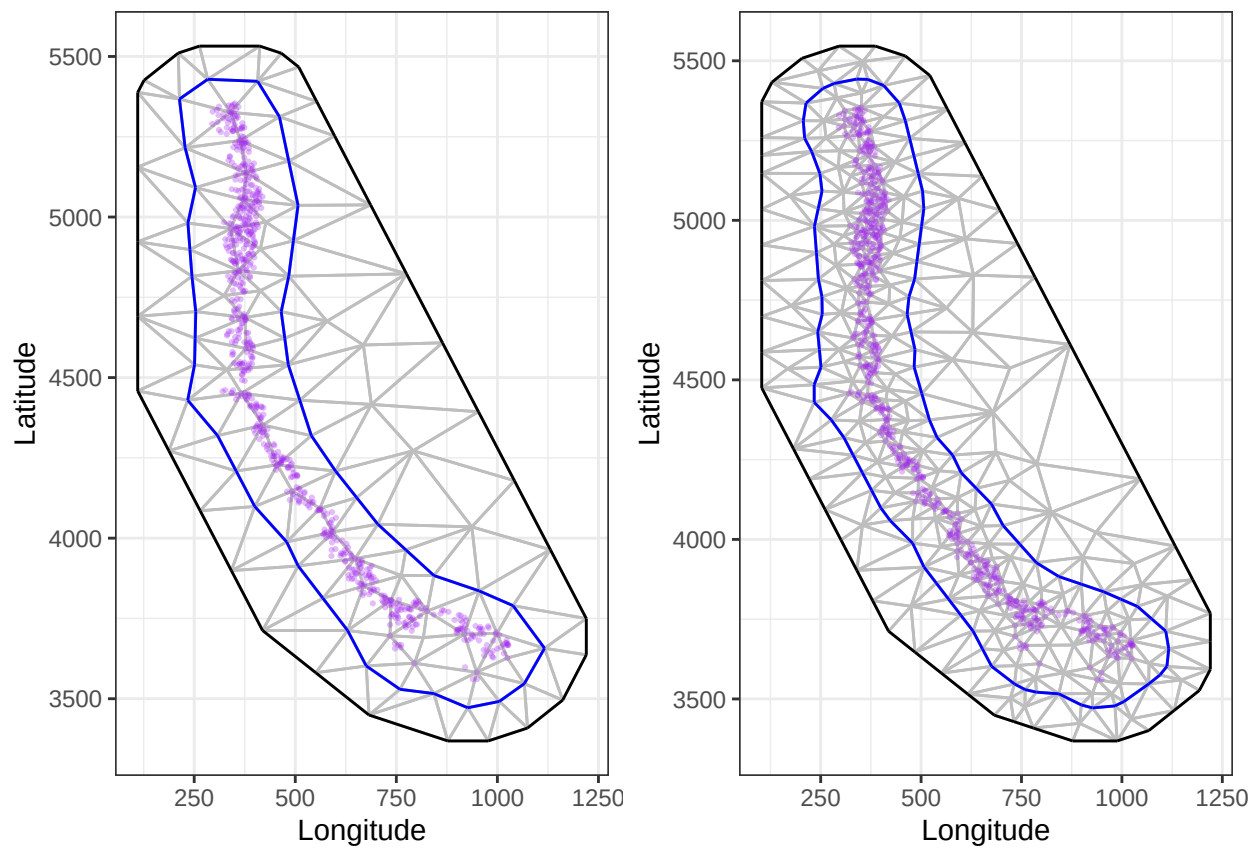


Figure 1: Example of two meshes constructed for the California Current Ecosystem. Cutoff values of 80 and 20km are used, translating into 110 and 394 knots (vertices) respectively.

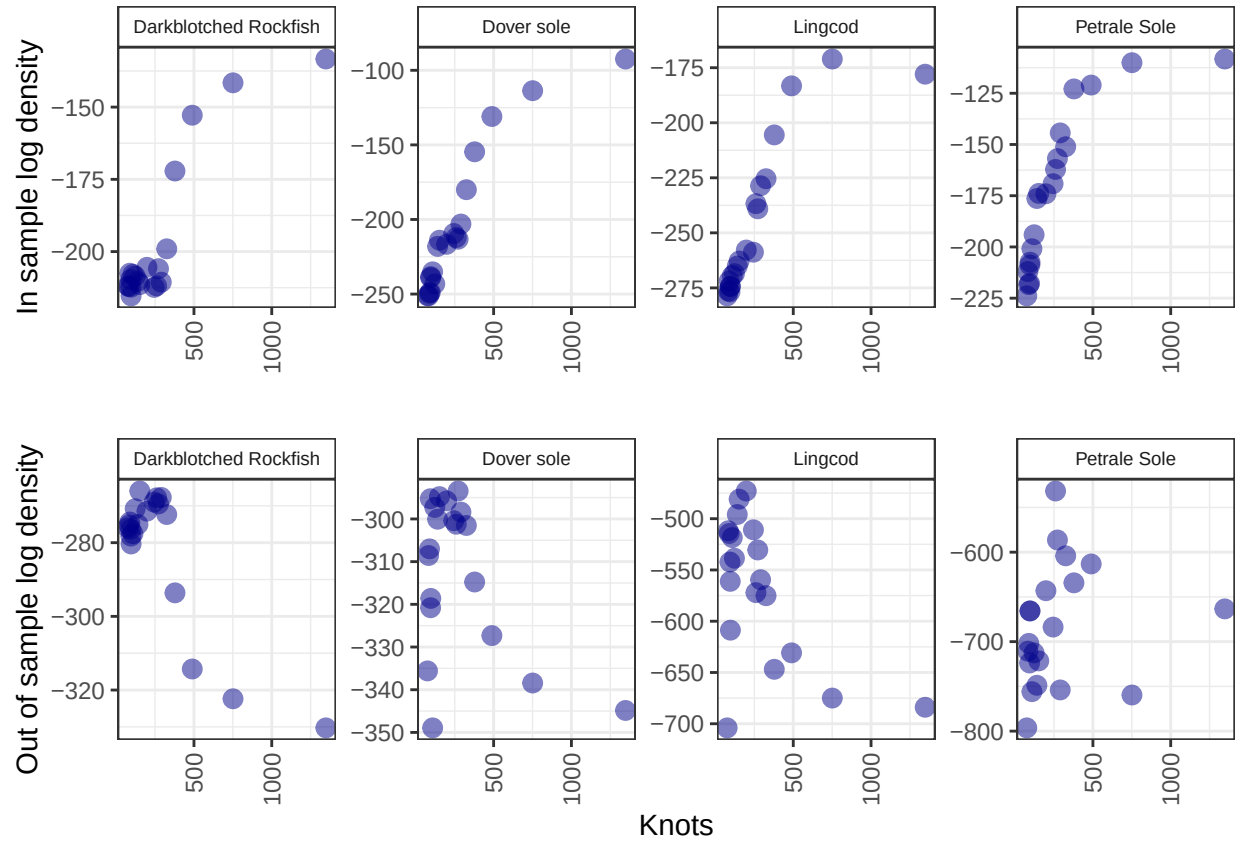


Figure 2: Binomial predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km)

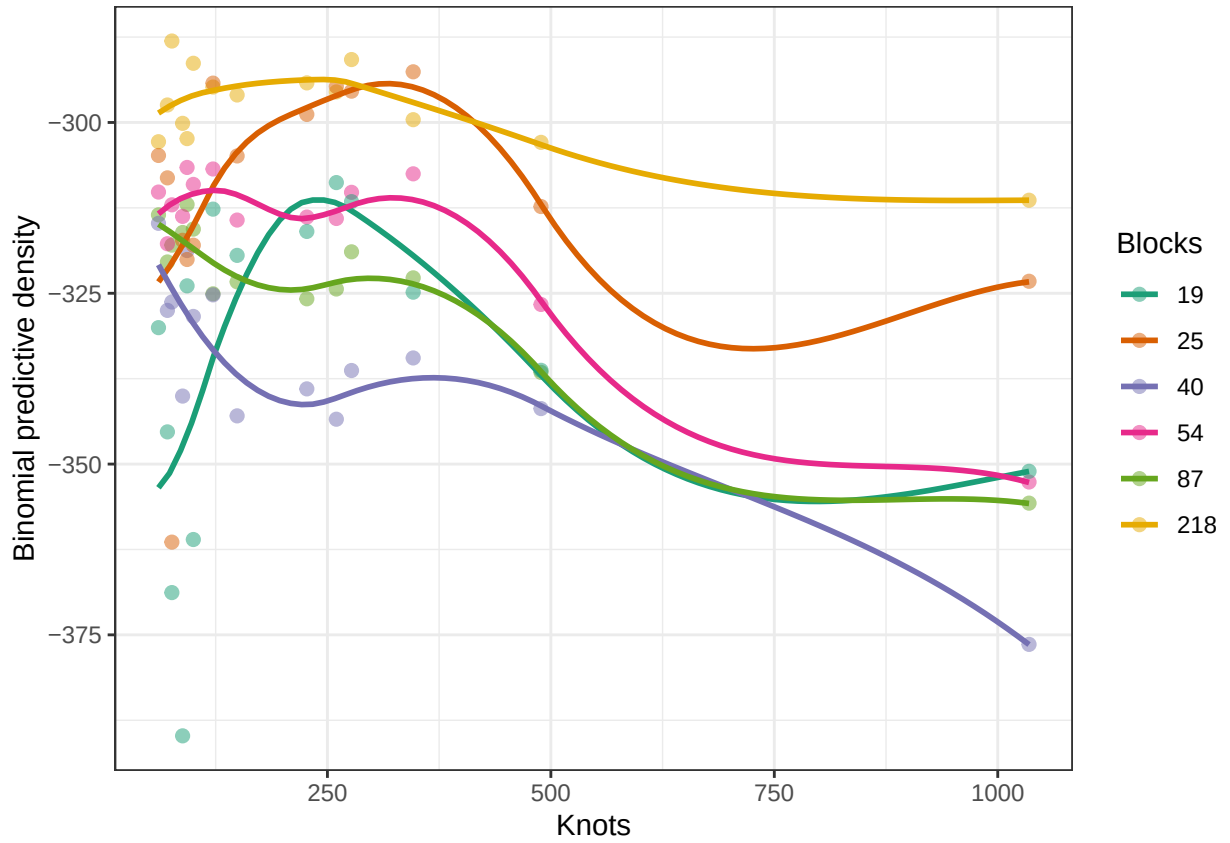


Figure 3: Binomial predictive density for Dover Sole, as a function of number of knots and number of spatial blocks used for 10-fold spatial cross-validation. Cutoff values between 5-135km are used across all scenarios.

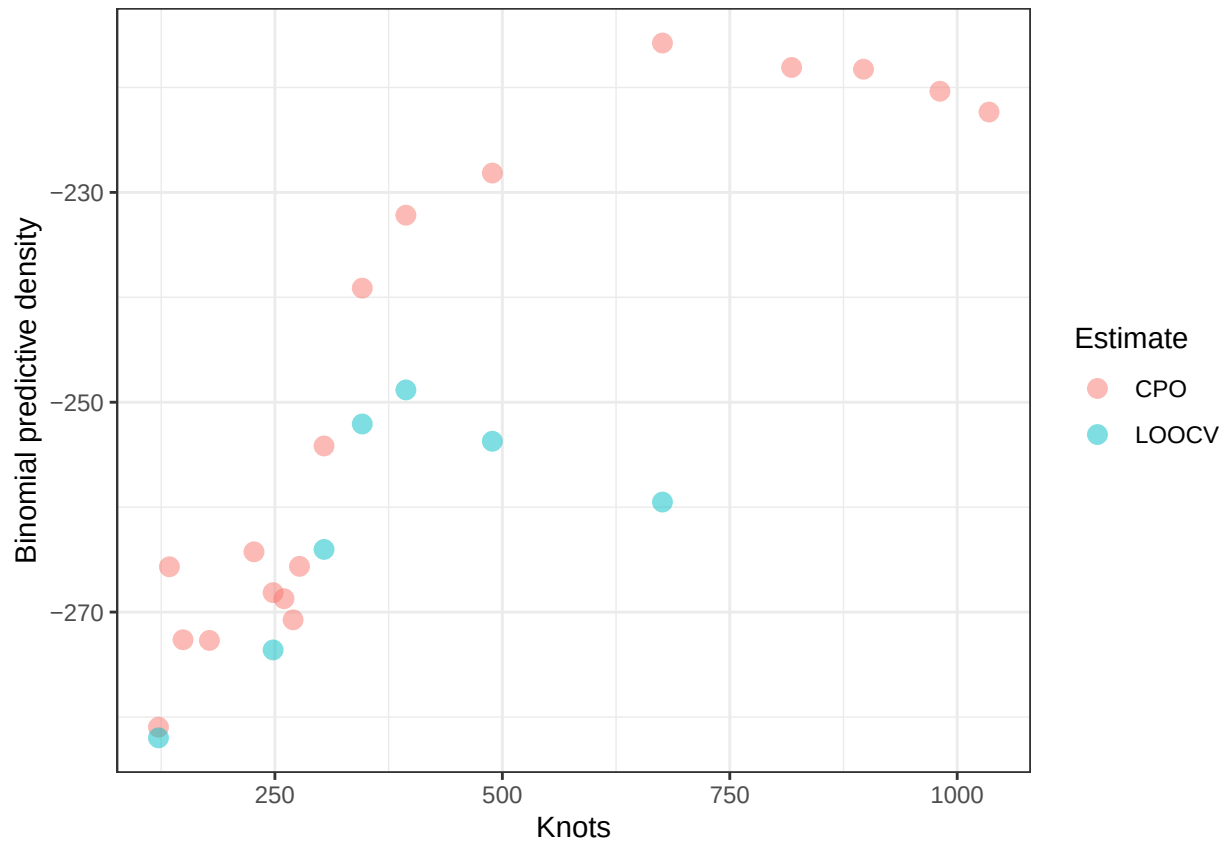


Figure 4: Comparison of CPO and LOOCV for the binomial presence-absence model of Dover sole. A wider range of cutoff values is shown for CPO (5-100km) than LOOCV (10-75km) to reduce computational time.

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281 {r fig-s2, echo=FALSE, fig.cap = "Global intercepts (and 95%
282 CIs) for four species from the WCBTS survey, using the Gamma
283 positive model, as a function of number of knots (using cutoff
284 values from 3-120km)", fig.pos="placeHere"} # source("figures/figure_S2_
285 # g1 #

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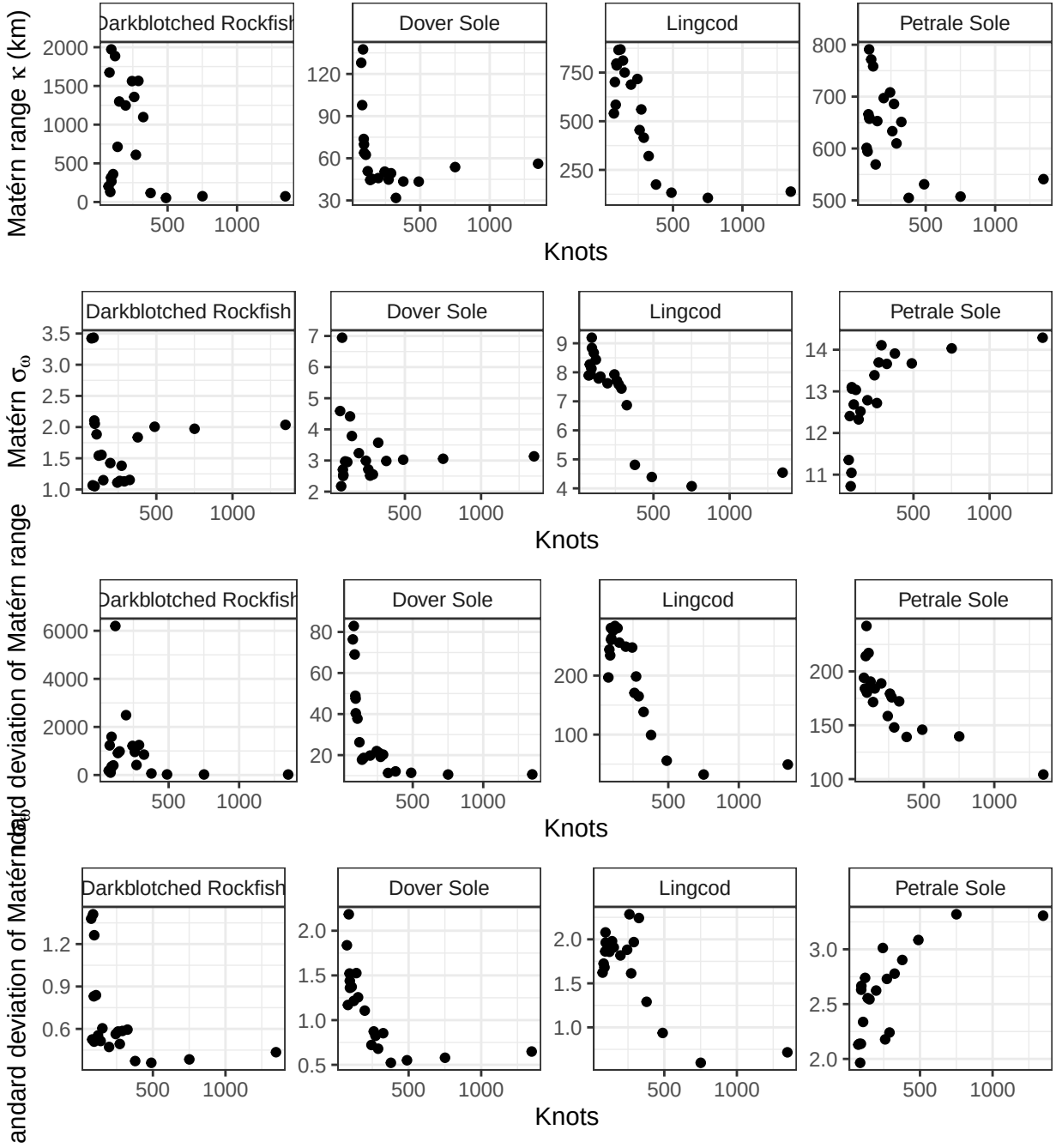


Figure 5: Estimates of the Matérn range and spatial scale, along with standard deviations for each parameter, across four species in the WCBTS survey. Cutoff values from 3-120km are shown.

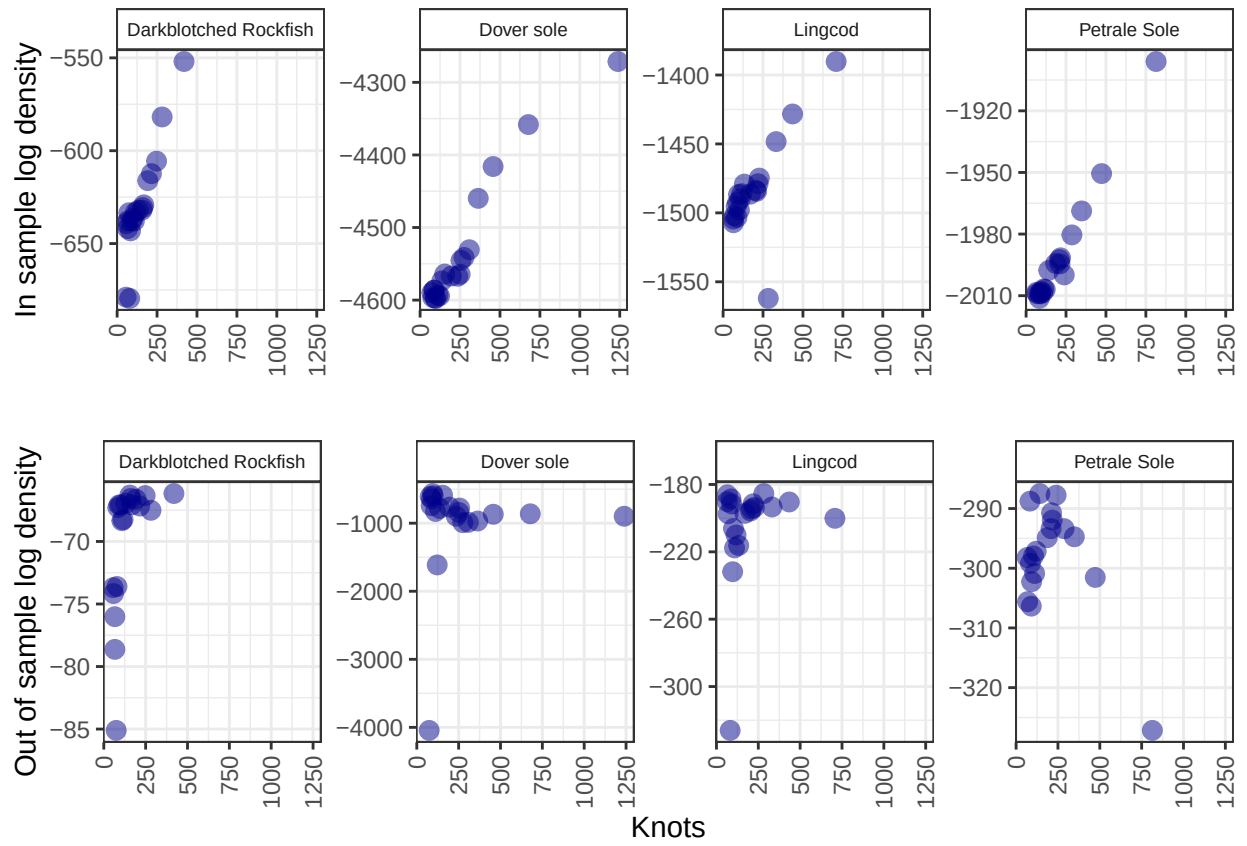


Figure 6: Gamma predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km)

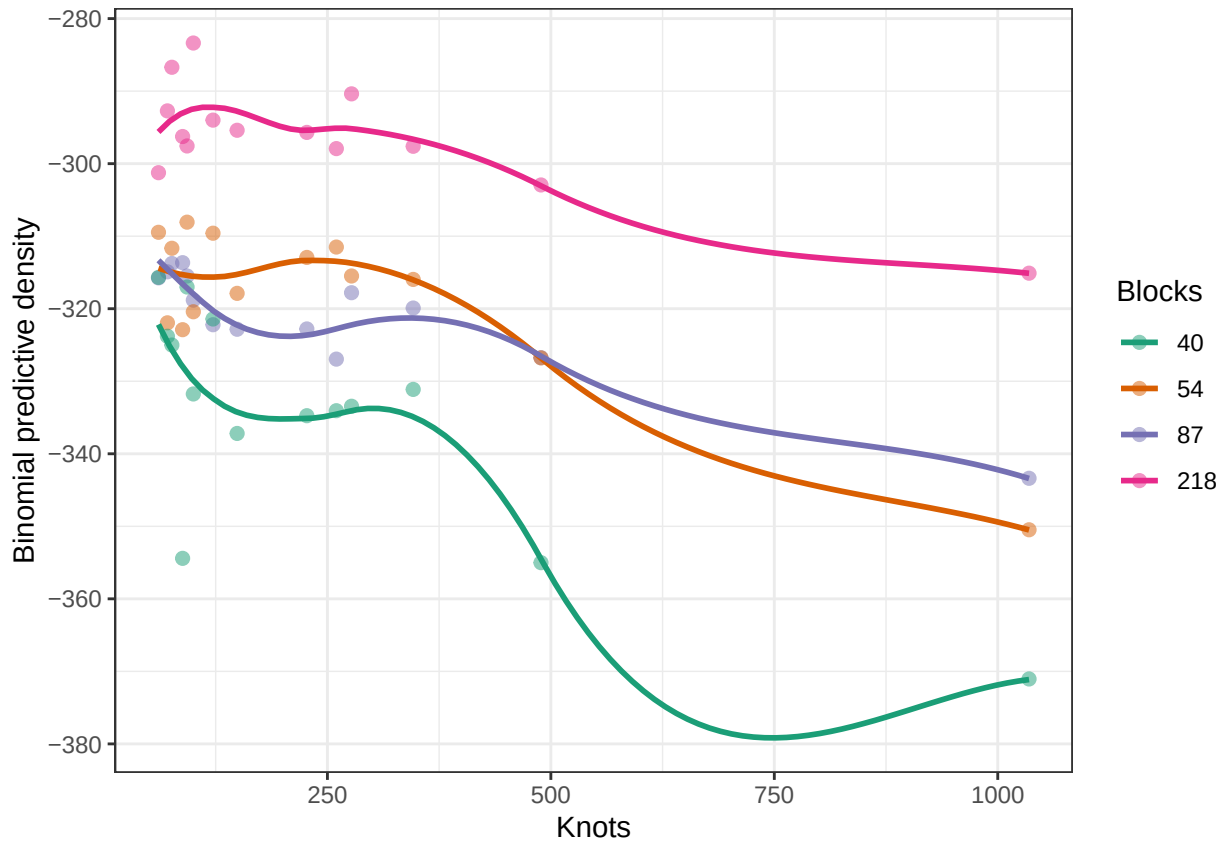


Figure 7: Binomial predictive density for Dover Sole, as a function of number of knots and number of spatial blocks used for 40-fold spatial cross-validation. Cutoff values between 5-135km are used across all scenarios.

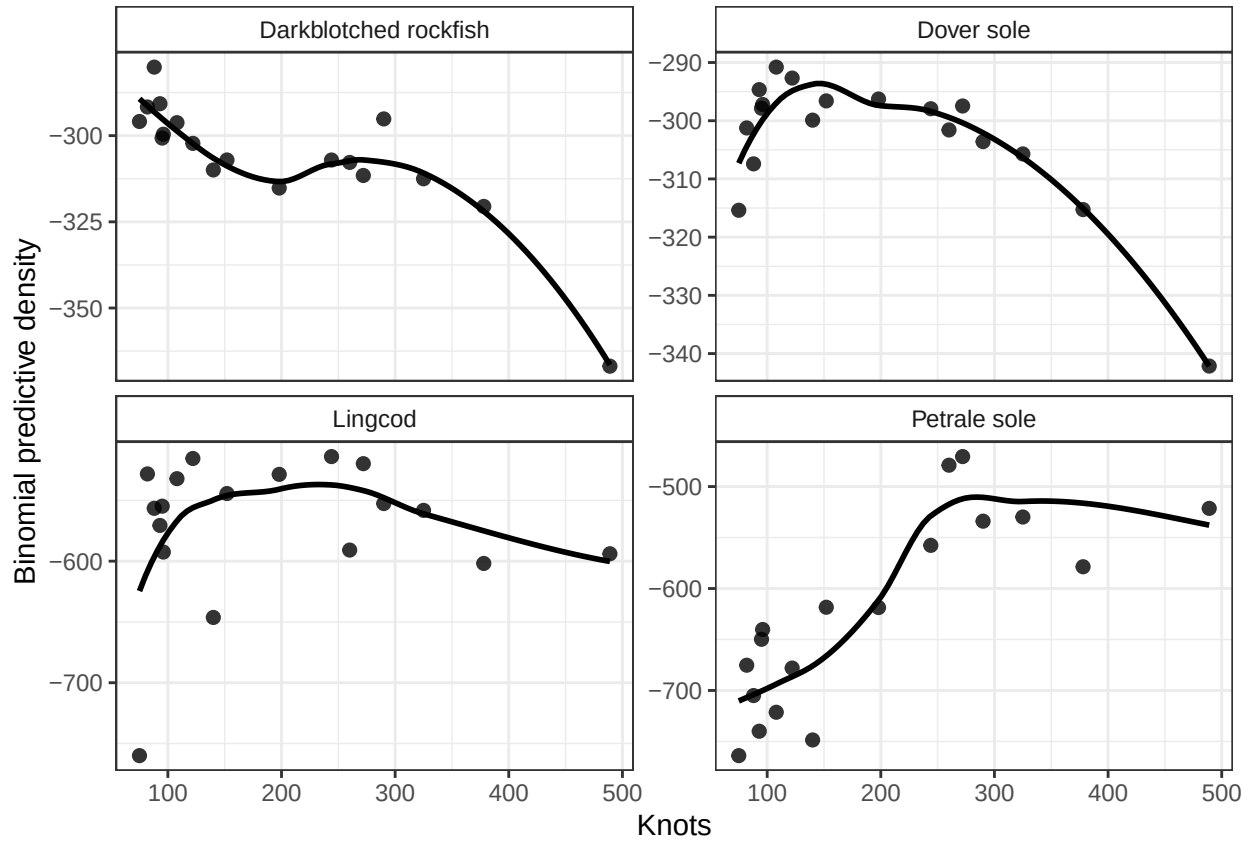


Figure 8: Binomial predictive densities for four species from the WCBTS survey, as a function of number of knots (using cutoff values from 3-120km). Estimation is done using Template Model Builder (TMB) instead of INLA.

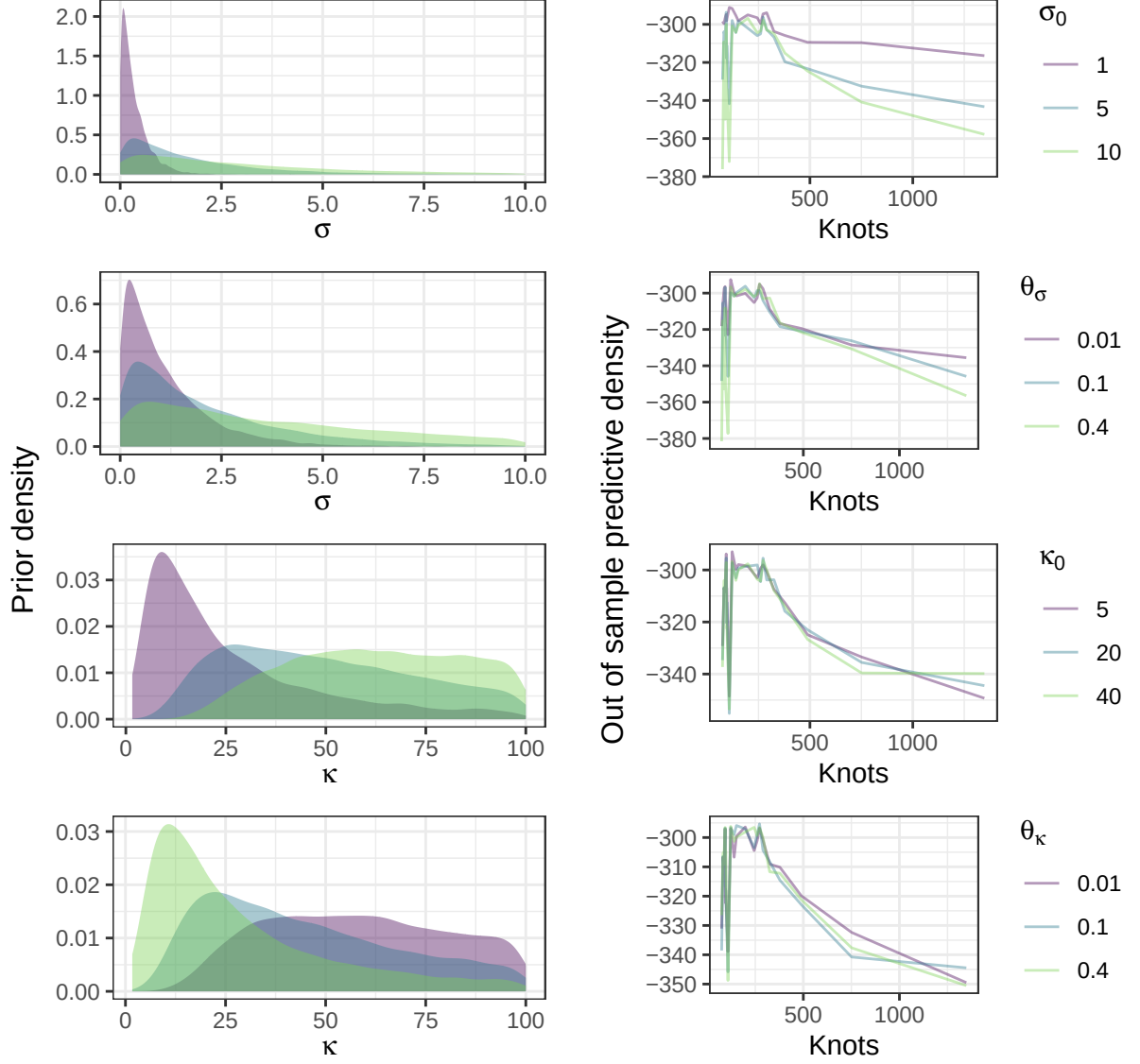


Figure 9: Prior sensitivity for Dover Sole presence-absence model, as a function of number of knots (using cutoff values from 3-120km) and hyperparameters of the Matern function.

References

- Anderson, S.C., Keppel, E.A. & Edwards, A.M. (2019b). A reproducible data synopsis for over 100 species of British Columbia groundfish. *DFO Can. Sci. Advis. Sec. Res. Doc.*, **2019/041**.
- Anderson, S.C., Keppel, E.A. & Edwards, A.M. (2019a). A reproducible data synopsis for over 100 species of British Columbia groundfish. *DFO Can. Sci. Advis. Sec. Res. Doc.*, **2019/041**.
- Anderson, S.C. & Ward, E.J. (2019). Black swans in space: Modeling spatiotemporal processes with extremes. *Ecology*, **100**, e02403.
- Bachl, F.E., Lindgren, F., Borchers, D.L. & Illian, J.B. (2019). Inlabru: An R package for Bayesian spatial modelling from ecological survey data. *Methods in Ecology and Evolution*, **10**, 760–766.
- Bakka, H., Rue, H., Fuglstad, G.-A., Riebler, A., Bolin, D., Illian, J., Krainski, E., Simpson, D. & Lindgren, F. (2018). Spatial modeling with R-INLA: A review. *WIREs Computational Statistics*, **10**, e1443.
- Banerjee, S., Carlin, B. & Gelfand, A. (2004). *Hierarchical Modeling and Analysis of Spatial Data*. Chapman & Hall/CRC.
- Banerjee, A., Dunson, D.B. & Tokdar, S.T. (2012). Efficient Gaussian process regression for large datasets. *Biometrika*, **100**, 75–89.
- Banerjee, S. & Fuentes, M. (2012). Bayesian modeling for large spatial datasets. *WIREs Computational Statistics*, **4**, 59–66.
- Datta, A., Banerjee, S., Finley, A.O. & Gelfand, A.E. (2016). Hierarchical Nearest-Neighbor Gaussian Process Models for Large Geostatistical Datasets. *Journal of the American Statistical Association*, **111**, 800–812.
- Forney, K.A., Ferguson, M.C., Becker, E.A., Fiedler, P.C., Redfern, J.V., Barlow, J., Vilchis, I.L. & Ballance, L.T. (2012). Habitat-based spatial models of cetacean density in the eastern Pacific Ocean. *Endangered Species Research*, **16**, 113–133.
- Fuglstad, G.-A., Simpson, D., Lindgren, F. & Rue, H. (2018). Constructing priors that penalize the complexity of gaussian random fields. *Journal of the American Statistical Association*, **114**.
- Garton, N., Niemi, J. & Carriquiry, A. (2020). Knot selection in sparse gaussian processes with a variational objective function. *Statistical Analysis and Data Mining: The ASA Data Science Journal*, **13**, 324–336.

- Godana, A.A., Mwalili, S.M. & Orwa, G.O. (2019). Dynamic spatiotemporal modeling of the infected rate of visceral leishmaniasis in human in an endemic area of Amhara regional state, Ethiopia. *PLOS ONE*, **14**, e0212934.
- Gómez-Rubio, V. (2020). *Bayesian inference with INLA*.
- Harrison, X.A., Donaldson, L., Correa-Cano, M.E., Evans, J., Fisher, D.N., Goodwin, C.E.D., Robinson, B.S., Hodgson, D.J. & Inger, R. (2018). A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ*, **6**, e4794.
- Held, L., Schrödle, B. & Rue, H. (2010). Posterior and Cross-validatory Predictive Checks: A Comparison of MCMC and INLA. *Statistical Modelling and Regression Structures: Festschrift in Honour of Ludwig Fahrmeir* (eds T. Kneib & G. Tutz), pp. 91–110. Physica-Verlag HD, Heidelberg.
- Huang, J., Malone, B.P., Minasny, B., McBratney, A.B. & Triantafyllis, J. (2017). Evaluating a Bayesian modelling approach (INLA-SPDE) for environmental mapping. *The Science of the Total Environment*, **609**, 621–632.
- Johnson, K.F., Thorson, J.T. & Punt, A.E. (2019). Investigating the value of including depth during spatiotemporal index standardization. *Fisheries Research*, **216**, 126–137.
- Keller, A.A., Wallace, J.R. & Methot, R.D. (2017). *The Northwest Fisheries Science Center’s West Coast Groundfish Bottom Trawl Survey: History, Design, and Description*. Seattle, WA.
- Kristensen, K., Nielsen, A., Berg, C.W., Skaug, H. & Bell, B.M. (2016). TMB: Automatic Differentiation and Laplace Approximation. *Journal of Statistical Software*, **70**, 1–21.
- Latimer, A.M., Banerjee, S., Jr, H.S., Mosher, E.S. & Jr, J.A.S. (2009). Hierarchical models facilitate spatial analysis of large data sets: A case study on invasive plant species in the northeastern United States. *Ecology Letters*, **12**, 144–154.
- Lindgren, F. & Rue, H. (2015). Bayesian Spatial Modelling with R-INLA. *Journal of Statistical Software*, **63**, 1–25.
- Lindgren, F., Rue, H. & Lindström, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: The stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **73**, 423–498.
- Miller, D.L., Burt, M.L., Rexstad, E.A. & Thomas, L. (2013). Spatial models for distance sampling data: Recent developments and future directions. *Methods in Ecology and Evolution*, **4**, 1001–1010.

342 Miller, D.L., Glennie, R. & Seaton, A.E. (2020). Understanding the Stochastic Partial Differential Equation
343 Approach to Smoothing. *Journal of Agricultural, Biological and Environmental Statistics*, **25**, 1–16.

344 Murase, H., Nagashima, H., Yonezaki, S., Matsukura, R. & Kitakado, T. (2009). Application of a generalized
345 additive model (GAM) to reveal relationships between environmental factors and distributions of pelagic
346 fish and krill: A case study in Sendai Bay, Japan. *ICES Journal of Marine Science*, **66**, 1417–1424.

347 Mutanga, O., Adam, E. & Cho, M.A. (2012). High density biomass estimation for wetland vegetation using
348 WorldView-2 imagery and random forest regression algorithm. *International Journal of Applied Earth
349 Observation and Geoinformation*, **18**, 399–406.

350 Osgood-Zimmerman, A. & Wakefield, J. (2021). A Statistical Introduction to Template Model Builder: A
351 Flexible Tool for Spatial Modeling.

352 Pettit, L.I. (1990). The conditional predictive ordinate for the normal distribution. *Journal of the Royal
353 Statistical Society: Series B (Methodological)*, **52**, 175–184.

354 R Core Team. (2021). *R: A language and environment for statistical computing*. R Foundation for Statistical
355 Computing, Vienna, Austria.

356 Righetto, A.J., Faes, C., Vandendijck, Y. & Ribeiro, P.J. (2020). On the choice of the mesh for the analysis
357 of geostatistical data using R-INLA. *Communications in Statistics - Theory and Methods*, **49**, 203–220.

358 Ross, B.E., Hooten, M.B. & Koons, D.N. (2012). An Accessible Method for Implementing Hierarchical
359 Models with Spatio-Temporal Abundance Data. *PLOS ONE*, **7**, e49395.

360 Rue, H., Martino, S. & Chopin, N. (2009). Approximate Bayesian inference for latent Gaussian models
361 by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B
362 (Statistical Methodology)*, **71**, 319–392.

363 Ruppert, D. (2002). Selecting the Number of Knots for Penalized Splines. *Journal of Computational and
364 Graphical Statistics*, **11**, 735–757.

365 Simpson, D.P., Rue, H., Martins, T.G., Riebler, A. & Sørbye, S.H. (2015). Penalising model component
366 complexity: A principled, practical approach to constructing priors. *arXiv:1403.4630 [stat]*.

367 Simpson, D., Rue, H., Riebler, A., Martins, T.G. & Sørbye, S.H. (2017). Penalising Model Component
368 Complexity: A Principled, Practical Approach to Constructing Priors. *Statistical Science*, **32**, 1–28.

369 Thorson, J.T. (2019a). Guidance for decisions using the Vector Autoregressive Spatio-Temporal (VAST)
370 package in stock, ecosystem, habitat and climate assessments. *Fisheries Research*, **210**, 143–161.

- Thorson, J.T. (2019b). Measuring the impact of oceanographic indices on species distribution shifts: The spatially varying effect of cold-pool extent in the eastern Bering Sea. *Limnology and Oceanography*, **64**, 2632–2645.
- Thorson, J.T. & Barnett, L.A.K. (2017). Comparing estimates of abundance trends and distribution shifts using single- and multispecies models of fishes and biogenic habitat. *ICES Journal of Marine Science*, **74**, 1311–1321.
- Tolimieri, N., Wallace, J. & Haltuch, M. (2020). Spatio-temporal patterns in juvenile habitat for 13 groundfishes in the California Current Ecosystem. *PLOS ONE*, **15**, e0237996.
- Valavi, R., Elith, J., Lahoz-Monfort, J.J. & Guillera-Arroita, G. (2019). blockCV: An r package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models. *Methods in Ecology and Evolution*, **10**, 225–232.
- Velasco, J.A. & González-Salazar, C. (2019). Akaike information criterion should not be a ‘test’ of geographical prediction accuracy in ecological niche modelling. *Ecological Informatics*, **51**, 25–32.
- Williamson, L.D., Scott, B.E., Laxton, M.R., Bachl, F.E., Illian, J.B., Brookes, K.L. & Thompson, P.M. (2021). Spatiotemporal variation in harbor porpoise distribution and foraging across a landscape of fear. *Marine Mammal Science*, **n/a**.
- Wilson, B. (2021). Evaluating the INLA-SPDE approach for Bayesian modeling of earthquake damages from geolocated cluster data.
- Wood, S.N. (2003). Thin plate regression splines. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, **65**, 95–114.
- Yue, Y.R., Simpson, D., Lindgren, F. & Rue, H. (2014). Bayesian adaptive smoothing splines using stochastic differential equations. *Bayesian Analysis*, **9**, 397–424.